Instructions for MMUQ Assignment 01

The first assignment is about finding the best model parameters for the HBV001a model to simulate the given hydrograph and model input data on Moodle. The assignment was discussed already in the class with the goals that need to be achieved. Here, a possible structure of the assignment is given without going into too much detail (which is your job).

To do so, multiple steps are needed such as defining the Objective Function (OF), setting up the HBV001a model in a form where it can be run by passing Parameter Vectors (PVs) to it, choosing an Optimization Scheme (OS), defining variables to track each sampled Parameter Vector (PV) and Objective Function Value (OFV) and finally presenting all the results.

The Optimization Scheme (OS):

There are thousands of OSs out there. We use "Differential Evolution (DE)" from SciPy. It requires the definition of an Objective Function (OF) and model Parameter Bounds (PRM_BDS, available on Moodle) along with all the data that HBV001a needs as input. While optimizing, DE samples many Parameter Vectors (PVs) and passes them to the OF along with the model and its inputs. The OF yields the Objective Function Value (OFV). DE then uses the OFV to determine if a sampled PV should be kept or discarded and finally yields the best performing one at the end. Depending on the OS, the form of the OF needs to be updated.

Defining the Objective Function (OF):

Traditionally, an OF needs to be minimized by the OS. The smallest value possible is 0. However, model efficiency metrics, such as NSE, need to have a value as close as possible to 1 i.e., smaller values of NSE are undesired. To convert an efficiency metric, similar to NSE, the Objective Function Value (OFV) is defined by subtracting them from their best possible value which in this case would be 1 – NSE. As the efficiency of the model needs to be calculated against a reference (observed values), the model under testing and the accompanying inputs need to be passed to the OF as well.

Tracking all sampled Parameter Vectors (PVs) and Objective Function Values (OFVs):

DE, for example, does not record everything that it does. Hence, we need to provide additional variables for keeping track of all the PVs and OFVs. This is simply done by using lists in the OF that save all the PVs and OFVs. Later when the OS has finished. Each PV (actually each parameter) can be plotted against its OFV. Similarly, the evolution of each parameter and the OFV can be plotted to see how OS reached the best value of each parameter and the OFV. Try to plot the best running OFV as well. This will reveal how often the best solution was updated.